

Caleb Pollreis

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EDUCATION

University of Manitoba

Bachelor of Science in Computer Engineering (Co-op)

- GPA: 3.87
- Internship Availability: Jan. 2027 - Aug. 2027

Winnipeg, Manitoba

Expected Graduation Date - April 2028

AUTONOMOUS SYSTEMS EXPERIENCE

UMSAE Formula Electric - Autonomous System Lead (DSO)

Jan. 2026 – Present

Winnipeg, Manitoba

- Leading 10+ members building an autonomous software stack to the **FSAE** driverless ruleset and timeline, with the goal of being driverless by 2028
- Secured **\$30k** in sponsored **LiDARs**/cameras and architected a **C++/ROS2** monorepo for the **NVIDIA Jetson Orin**
- Architected a **C++/ROS2** monorepo targeting the **NVIDIA Jetson Orin** platform, establishing build infrastructure and project structure for the autonomous software stack

PTx Trimble - Perception Engineering Intern

May 2026 – Present

Winnipeg, Manitoba

- Trained a **ViT** student model on a **Jetson Orin** via **knowledge distillation** from a **CLIP** teacher, cutting size 33% and inference time $\sim 2\times$ while raising accuracy 15%
- Built **ROS** fault-detection bag triggers and a **C++/Cloudini** pipeline to decode point cloud and **H.264** data

MacDon Industries Ltd. - Software Developer Intern

Jan. 2025 – Aug. 2025

Winnipeg, Manitoba

- Built a **computer vision** grid algorithm using a **deep learning** model to detect crop conditions and automate combine operations
- Scaled a **ROS** multi-camera vision pipeline and added **CAN J1939** cut-height control on **NVIDIA Jetson**

UMSAE Formula Electric - Firmware System Lead

Aug. 2024 – June 2026

UMSAE Formula Electric - Software Team Member

Sept. 2022 – Aug. 2024

UMSAE Executive Team - Web Director

June 2024 – Oct. 2025

St. Ignatius School - After-School Program Leader

Sept. 2022 – Jan. 2023

Math Tutor - High School

Jan. 2020 – June 2022

PROJECTS

Rocket League BOT | 🐙 CPollreis/rocket-league-bot

- Offline bot trained with **reinforcement learning** (**RLGym**, **PyTorch**) using custom reward shaping in a mocked **Unreal Engine** environment

BMO - An Embodied AI Agent | 🐙 CPollreis/bmo-agent

- Local multimodal AI agent running a quantized **Qwen 3 LLM** on a **Raspberry Pi 5** with **Whisper** voice I/O and live camera context

NAS Storage System | 🐙 CPollreis/nas-storage

- Built a **NAS** on commodity hardware with **RAID** redundancy and a **Jellyfin** media server for local and remote streaming

TECHNICAL SKILLS

Programming Languages: C++, C, Python, Verilog, Java, JavaScript, R

Machine Learning: PyTorch, ONNX, NVIDIA TensorRT, CUDA Toolkit, Netron, Dataset Synthesis

Robotics & Computer Vision: ROS2, OpenCV, Eigen, Cloudini

Cloud: AWS (EC2, S3, SageMaker)

Build & Dev Tools: Docker, CMake, GTest, LLVM, clang-tidy, cpplint

Embedded & Protocols: FreeRTOS, STM32Cube, Raspberry Pi, CAN, SPI, UART/USART, I2C, TCP/IP

Hardware & Fabrication: CAD (SolidWorks), 3D Printing

AI Tools: Claude, Codex, Cursor

Version Control & Collaboration: Git, GitHub, Bitbucket, Jira

AWARDS AND ACHIEVEMENTS

Scholarships & Bursaries: Ron and Dorothy Britton Design Engineering Bursary (2x), W.J. Christie Memorial Bursary in Engineering, Guertin Centennial Scholarship, University of Manitoba Student Union Bursary

Professional Development: AWS Cloud Practitioner Workshop